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Context-Conditioned Causal Alignment for Scientific Literature (CCST)

Separating Cross-Domain Analogies, True Contradictions, and Contextual Heterogeneity at Corpus Scale

TEAM 16 • PROJECT MID-TERM PRESENTATION

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PROBLEM DEFINITION

The Two Critical Blind Spots in Scientific Search

BLIND SPOT 1 Disjoint Vocabularies Conceal Analogies

Biology / Epidemiology: Models respiratory viral propagation using Susceptible-Exposed-Infectious-Recovered (SEIR) network clustering.

Computer Systems: Models self-replicating worm malware across software-defined network (SDN) switches using identical graph dynamics.

→ Dense Embeddings (SBERT, Contriever) score them as completely unrelated due to zero lexical overlap (0% Analogy Recall)!

BLIND SPOT 2 Superficial Opposites Cause False Contradictions

Paper A (Clinical Trial): Calcium channel blocker reduces systemic vascular resistance in *human adults* at *therapeutic doses*.

Paper B (Toxicology): Calcium channel blocker induces paradoxical reflex vasoconstriction in *murine models* under *toxic overdoses*.

→ Standard NLI / LLMs focus only on "reduces" vs "increases", generating a catastrophic 40%+ False-Contradiction Error Rate!

PROPOSED SOLUTION

Our Proposed 5-Way Scientific Relation Taxonomy

1. AGREEMENT

Both papers report congruent causal effects under compatible physical context and population.

2. CROSS-DOMAIN ANALOGY

Isomorphic causal mechanism graph topology expressed across disjoint scientific vocabularies.

3. CONTRADICTION CANDIDATE

Opposing causal polarity (±1 vs –1) under verified **identical material context**.

4. CONTEXTUAL HETEROGENEITY

Opposing outcome directions explained by divergent physical scope (species, dose, environment, or time horizon).

5. INSUFFICIENT COMPARABILITY

Incompatible core interventions or insufficient textual evidence to evaluate mechanism alignment.

MILESTONE 1 COMPLETED

2,054 Real Scientific Documents Ingested from Live APIs

500

PUBMED BIOMEDICAL

NCBI E-Utilities API (Oncology, Pharmacology, Cardiology)

496

ARXIV CS.CR

Computer Security (Malware, Intrusion, Watermarking)

480

ARXIV CS.NI

Networking (Congestion, Datacenter Fabrics, BGP)

478

ARXIV CS.SY / EESS.SY

Control Systems (Feedback, Stabilization, Swarms)

PROPOSAL QUALITY FILTERS ENFORCED

Strict English validation (≥85% ASCII check) • Minimum 50-word abstract length • Exclusion of retracted and withdrawn papers • Zero duplicates via normalized MD5 title fingerprints • Full preservation of real PMIDs, arXiv IDs, DOIs, and author lists.

MILESTONE 2 COMPLETED

240-Pair Four-Stratum Ground-Truth Benchmark

Stratum 1: Controlled Mutations (60 Pairs)

Objective node, edge, and sign labels by construction: 15 vocabulary mutations • 15 polarity sign flips • 15 physical context shifts • 15 subgraph noise injections.

Stratum 2: Cross-Domain Analogies (60 Pairs)

Pairs exhibiting isomorphic causal topologies across disjoint vocabularies (e.g., epidemic SIR models ↔ network worm malware).

Stratum 3: Natural Literature Pairs (60 Pairs)

Naturally occurring scientific papers annotated for agreement, true contradiction, and contextual heterogeneity.

Stratum 4: Adversarial Hard Negatives (60 Pairs)

Quoted-not-endorsed claims, simulation vs. observation, missing context dimensions, and lexically similar but structurally opposite claims.

HIGH INTER-ANNOTATOR RELIABILITY

Independent triplicate annotations across natural pairs before consensus adjudication.

Krippendorff's α = 0.767

MILESTONE 3 COMPLETED

Multi-Mechanism Extraction with Character-Span Provenance

Equation (1) Mechanism Claim Representation

$$q_{im} = (C, P, A, M, Y, \tau, \delta, \eta, u, p)_{im}$$

C: Material Context • P: Population/System • A: Intervention • M: Mediators • Y: Outcome • δ: Polarity (±1, 0) • η: Epistemic Status • $p = (start, end)$: Character Provenance Span.

CORPUS-SCALE EXTRACTION AUDIT

- **2,054** Real Papers Ingested
- **16,638** Causal Mechanisms Extracted (avg 8.1 / doc)
- **11,945** Graph Nodes compiled
- **35,057** Typed Directed Edges compiled
- Throughput: **965.4 docs/sec** on ASU SOL

100% PROVENANCE GROUNDING

Every mechanism claim retains the exact slice of text from the original abstract:

```
chars [145–230]: "...Atorvastatin (80 mg daily)
demonstrated a 48 percent reduction in LDL–C..."
–> Verified 100% character precision
```

MILESTONE 4 COMPLETED

Baseline Evaluations Expose the Core Problem

Baseline System	Type	Macro F1	False-Contradiction Rate	Analogy Recall
BM25 Retrieval [11]	Classical Lexical	14.8%	**38.0%**	0.0%
TF-IDF + LDA [12]	Topic Model	18.2%	**34.0%**	5.0%
Sentence-BERT [9]	Dense Embedding	24.5%	**42.0%**	0.0%
Contriever [10]	Contrastive Dense	22.1%	**44.0%**	0.0%
Direct LLM / NLI Judge	Cross-Encoder NLI	21.4%	**40.0%**	30.0%

- **Severe Failure: All standard NLP / IR models produce a 34%–44% False-Contradiction Rate on context-shifted literature!**
- **Vocabulary Siloing: SBERT, Contriever, and BM25 achieve 0.0% analogy recall because words in different disciplines do not match.**

PROPOSED SOLUTION

CCST: Context-Conditioned Signed Partial Transport

Equation (2) CCST Alignment Objective

$$\mathcal{D}_{\mathrm{CCST}}(G_i, G_j) = \min_{T \geq 0} \left[\sum M_{uu'} T_{uu'} + \alpha \sum L_{uv,u'v'} T_{uu'} T_{vv'} + \lambda_u U(T) + \lambda_g \Phi_{\mathrm{gate}}(G_i, G_j) \right]$$

1. PARTIAL TRANSPORT

$$\lambda_u U(T) = 2\lambda_u (1 - \mathbf{1}^\top T \mathbf{1})$$

Allows peripheral or irrelevant claims to remain unaligned without corrupting the core mechanism match.

2. SIGNED VS. UNSIGNED

$$\Delta_{\mathrm{sign}} = D_{\mathrm{signed}} - D_{\mathrm{unsign}}$$

Isolates true directional polarity flips (+1 vs -1) from underlying structural graph topology.

3. CONTEXT GATING

$$\Phi_{\mathrm{gate}}(G_i, G_j)$$

Penalizes contradiction declarations when species, dosage, or environment differ.

VALIDATION STATUS

Prototype Validation & Proof-of-Concept Success

PROOF-OF-CONCEPT RESULT 1

Context Gating Eliminates False Contradictions

- Evaluated across all 50 context-shifted benchmark pairs
- Pairs misclassified as contradictions by CCST: **0 / 50**
- False-Contradiction Error Rate: **0.00%** (vs. 40%+ on baselines!)
- Confirms Theorem 2: Context gating successfully blocks false contradictions.

PROOF-OF-CONCEPT RESULT 2

100% Analogy Recall via Unsigned View

- Discovers structural isomorphisms across disjoint fields
- Epidemic SEIR models ↔ SDN switch malware scanning: **D_unsign = 0.106**
- Cross-domain analogy recall: **100.0%** (vs. 0% for SBERT/BM25)

OUTPUT DELIVERABLE

Evidence-Linked Disagreement Certificate

Automatically outputs minimal conflicting subgraphs with character-level text spans, explaining whether divergence is a `CONTRADICTION_CANDIDATE` or `CONTEXTUAL_HETEROGENEITY`.

SCALABILITY ARCHITECTURE

Tri-Index Sparse Candidate Retrieval Engine

Solving the O(N²) Bottleneck

Comparing 2,054 papers brute-force requires **1,999,000 exact alignments**. We prune the search space to an empirical $O(N \log N + N \cdot k)$ via a union of 3 sparse indexes:

I_SEM: SEMANTIC

FAISS dense inner-product index over normalized 384-d Sentence-BERT vectors (top-$k_s = 15$).

I_GRAPH: STRUCTURAL

2-iteration Weisfeiler-Lehman (WL) graph hashing index over unsigned topologies (top-$k_m = 15$).

I_CTX: CONTEXTUAL

Inverted bitvector index filtering compatible physical scope dimensions (top-$k_c = 10$).

Search Space Reduction

1,999,000 → 80,000 candidate pairs (**25.0× speedup**).

GPU Indexing Throughput

2,054 graphs indexed in 0.23 seconds (**8,857 graphs/sec**).

TEAM GOVERNANCE

Primary Ownership & Division of Work (Table III)

Team Member	Primary Responsibilities	Mid-Term Status
Shravan Krishna Vijay	Corpus acquisition (PubMed/arXiv APIs), preprocessing, BM25 / TF-IDF baselines.	COMPLETED
Sri Harshith Goli	CCST mathematical formulation, Frank-Wolfe partial OT solver, signed/unsigned tests.	COMPLETED
Subhash Shivaprasad	Multi-mechanism extraction engine, character span provenance, context schemas.	COMPLETED
Tarun Kumar Bantupalli	Sparse ANN indexes (FAISS + WL), candidate scaling measurements, calibration.	COMPLETED
Vedang Vasant Avaghade	Benchmark coordination, statistical evaluation, certificates & result presentation.	COMPLETED

*All members participated in independent benchmark annotation ($\alpha = 0.767$), code review, and presentation preparation.

TIMELINE & NEXT STEPS

Roadmap to Final December Submission

Early November

- Systematic hyperparameter grid search ($\alpha, \lambda_u, \lambda_g$)
- Calibrated probability thresholds optimization on dev split

Mid November

- Full held-out test evaluation with 1,000 bootstrap resamples
- Comprehensive ablation studies on isolated components

Late November

- Full-corpus scalability benchmarks on ASU SOL A100 GPU
- Refinement of Disagreement Certificate UI & visualization

December

- Final technical report & code repository release
- Conference paper manuscript draft for submission (SIGKDD / ACL)

Thank You! Any Questions?